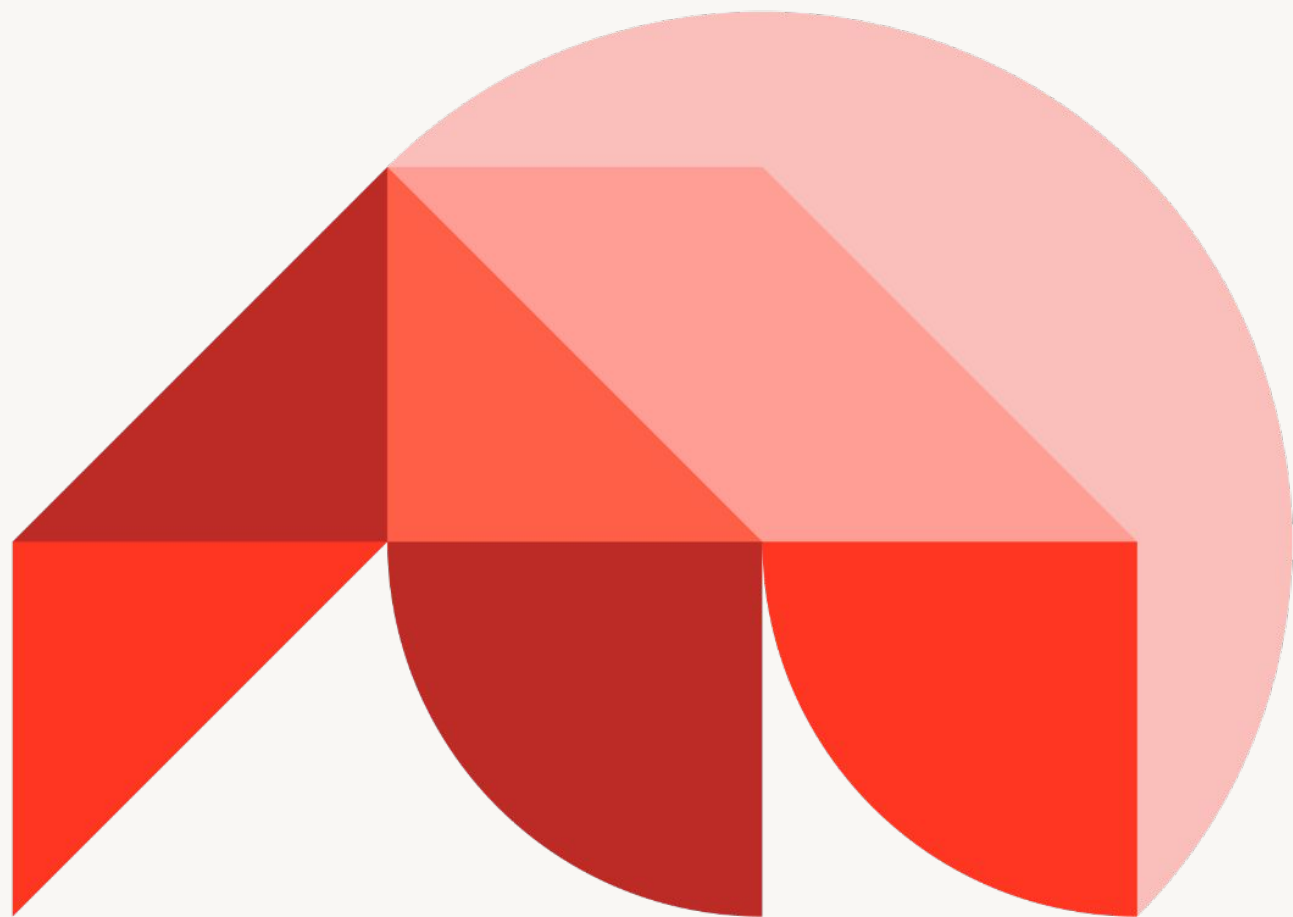
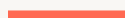




AI/BI Technical Overview

Intelligent analytics for everyone



AI/BI Genie – Core Concepts

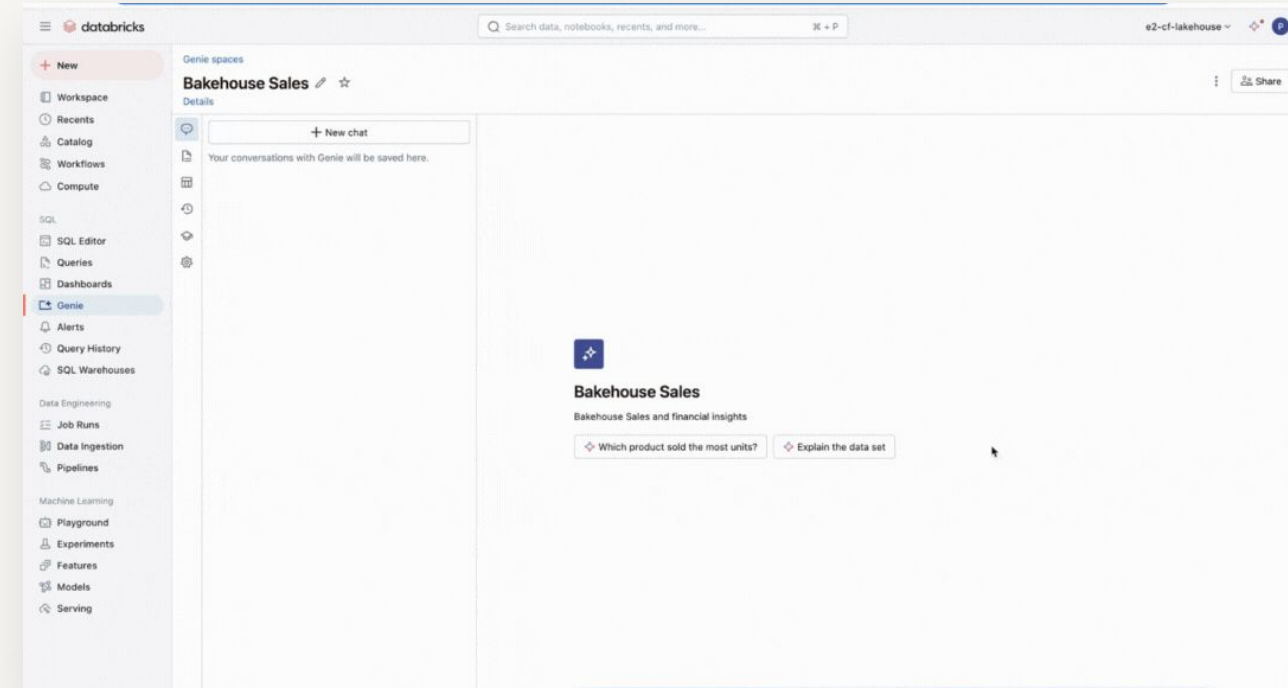


AI/BI Genie

Assistant with context knowledge

Allow practitioners to get insights without needing any pre-existing knowledge of your data layer

- **Self-service Q&A**—go beyond dashboards and converse directly with your data
- **AI with Data Intelligence**—accurate answers from AI with unique knowledge of your data that improves through user feedback
- **Governed and secure**—fully governed interactions that adhere to your policies established in Unity Catalog



AI/BI Genie – Semantics



Leverage Metric Views in Genie

Metrics provide determinism while tables augment with flexibility

What it is

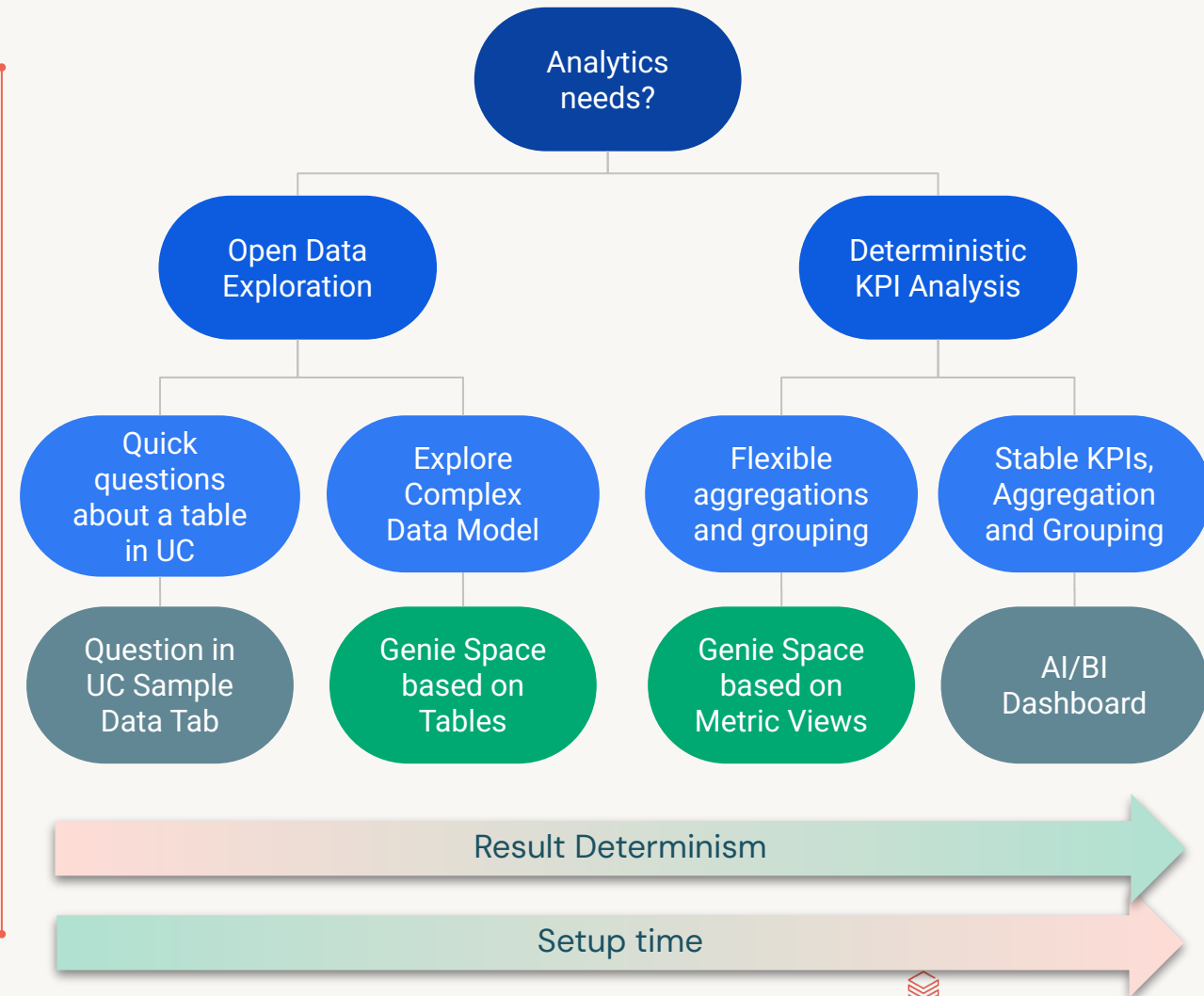
- Use Metric views (reusable KPIs through measures and dimensions) for business analysis, augment it with tables for open data exploration

Why it matters

- Genie will have more **deterministic answers** with Metric Views
- Generated queries will be simpler to interpret and validate

Key Benefits/Use Cases

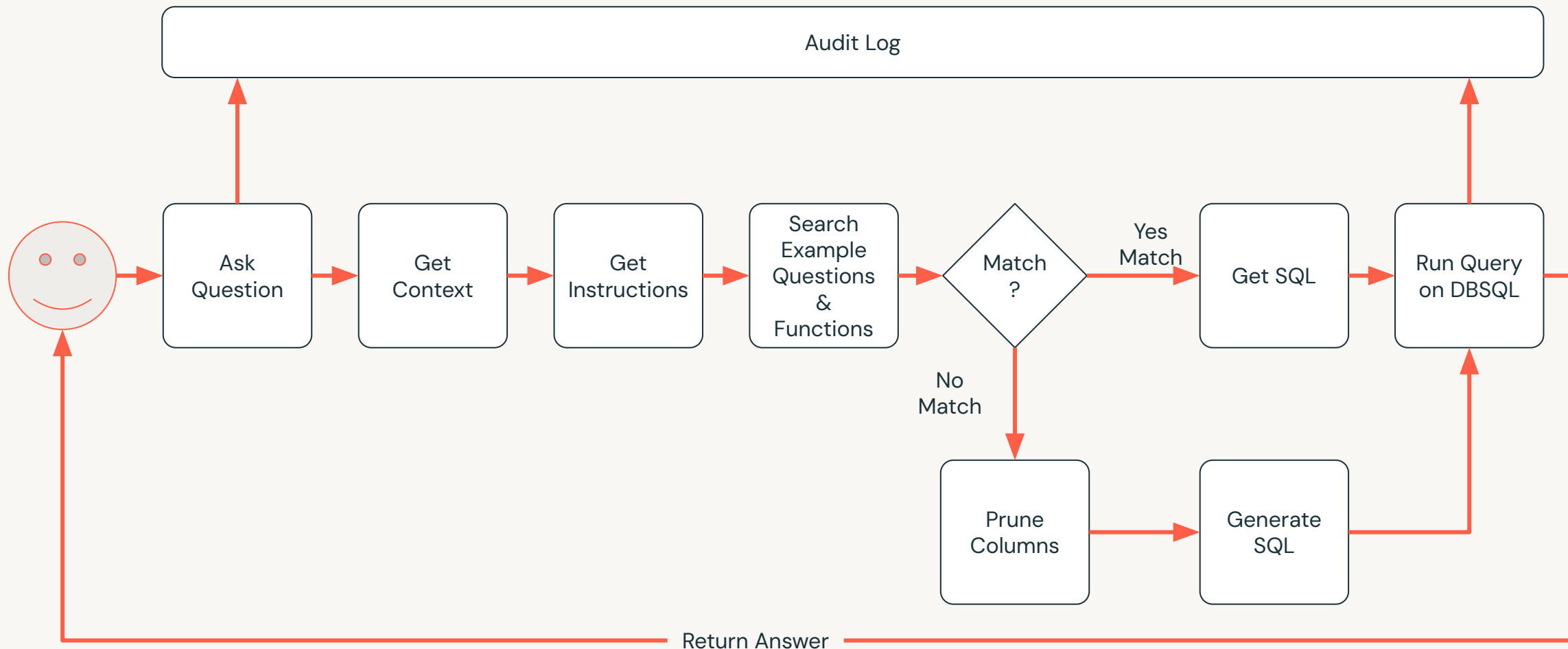
- **Improve Trust** for end-users
- Better reusability of business KPI definitions across spaces and use cases
- Easier maintenance in the long run (one central definition)



AI/BI Genie Architecture



AI/BI Genie Architecture

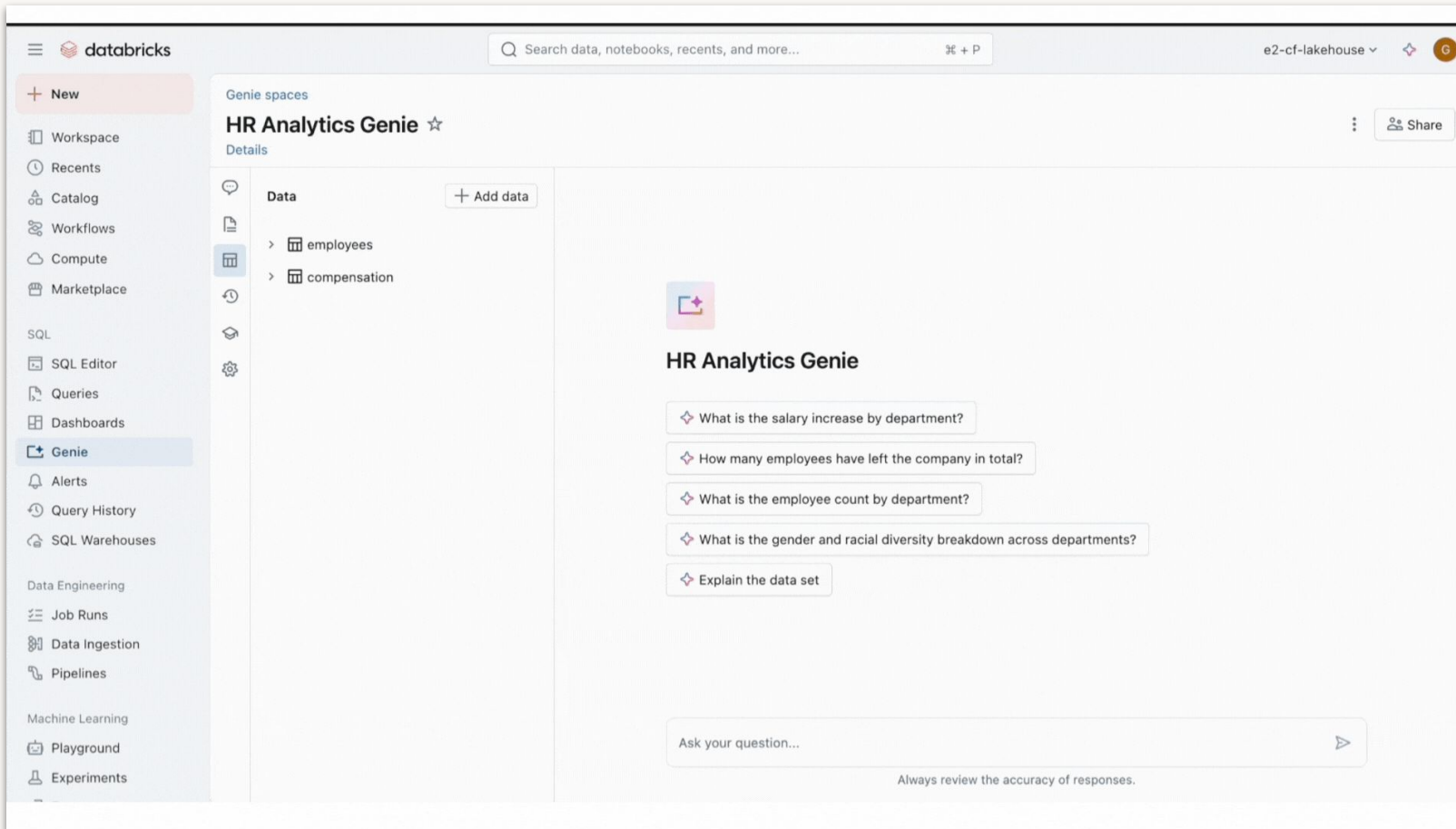


AI/BI Genie Setup and Lifecycle



Building your Genie Space

Start with well-documented tables



- Begin with UC tables/views and add meaningful comments to columns and table descriptions for clarity
- Clearly define primary keys and foreign keys for better context understanding

AI/BI Genie – Learning and Improving



Knowledge Mining

Learning semantics from *existing* platform usage

What it is

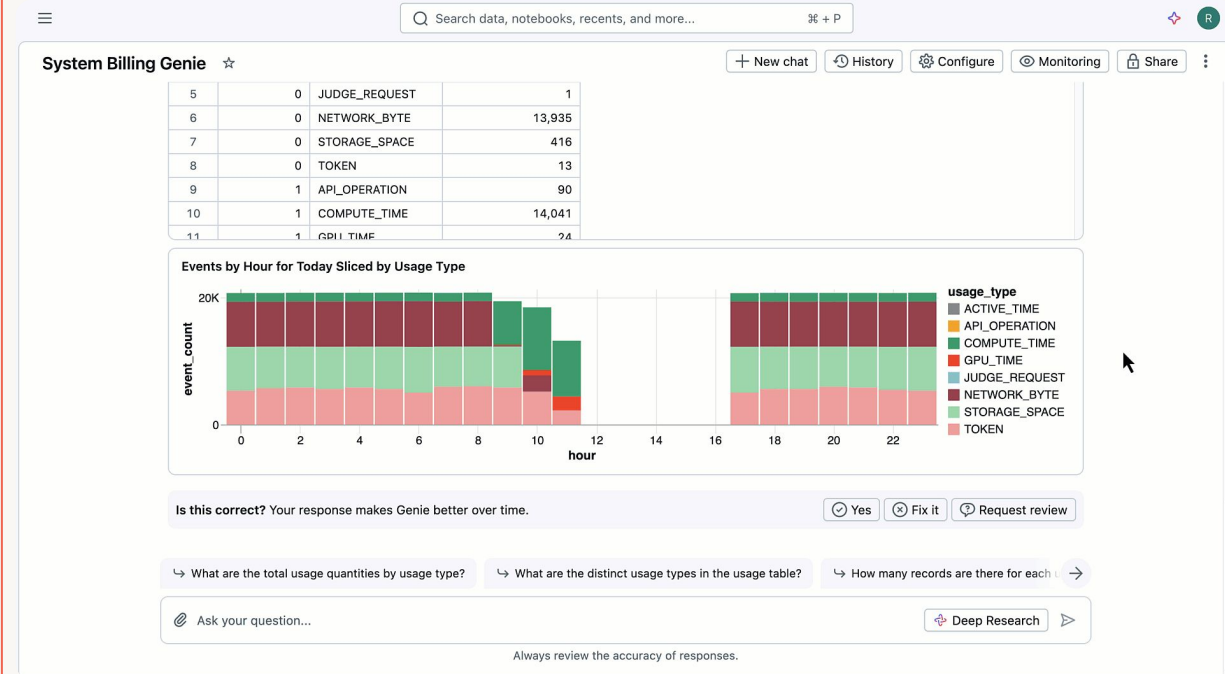
- Discovery and aggregation of relevant information from a variety of data sources across your environment; Genie uses this to surface both technical and business context

Why it matters

- Enables Genie to answer smarter questions

Key Benefits/Use Cases

- Metadata ingestion from Unity Catalog (table names, column descriptions, PK/FK relationships, lineage, access policies)
- General Instructions
- Example SQL Queries and Functions
- Genie builds a semantic map of your data landscape



Knowledge Extraction

Learning semantics from *new* Genie usage

What it is

- Transforms mined information into actionable knowledge, by merging metadata with business context for a unified understanding

Why it matters

- Consistent, accurate, and explainable insights tailored to your organization's data and terminology

Key Benefits/Use Cases

- Reduces confusion, boosts adoption, and builds trust in AI/BI

The screenshot displays the Genie interface, which is used for generating SQL queries from natural language instructions. It features three main panels: Data, Instructions, and Settings.

Left Panel (Data): Shows a SQL query for marketing tactics. The query is as follows:

```
1 WITH filtered_attribution AS (  
2   SELECT  
3     engagement_id,  
4     SUM(mql) AS mql  
5   FROM  
6     'marketing' AS mkt  
7   WHERE  
8     fiscal_year = 2023  
9     AND mql > 0  
10  GROUP BY  
11    engagement_id  
12 )  
13 joined_data  
14 SELECT  
15   e.engagement_id, m.mql
```

Middle Panel (Joins): A table showing the relationships between different tables and how they are joined together.

Left table	Right table	Condition
attributio	contacts	'attribution'.'a
engagem	contacts	'engagements'.'a
attributio	engagem	'attribution'.'e
outcome:	contacts	'outcomes'.'audi
attributio	outcome:	'attribution'.'co...
attributio	accounts	'attribution'.'ac...
consump	accounts	'consumption'.'di...

Right Panel (General Instructions): A section for adding general instructions on how you want Genie to behave. The instructions include:

- FY = Fiscal Year
- Region = business unit = BU = business_unit
- Comms, Media & Entertainment and CME both mean to filter where account_vertical = "CME"
- "Federal Government or Public Sector" means to filter where account_vertical = "PubSec & Fed"
- "AMER" business unit means to include: "AMER", "AMER Enterprise & Emerging" and "AMER Industries"
- For "delivered emails", "sent emails", or "opened emails" - do not include a filter where is_attr_engagement = true
- For all other queries against the engagement table, include this filter: WHERE is_attr_engagement true
- This is very important: When looking at

The interface also includes a "Save" button at the bottom right of the General Instructions panel.



User Feedback

Thumbs Up/Thumbs Down and Request Review

What it is

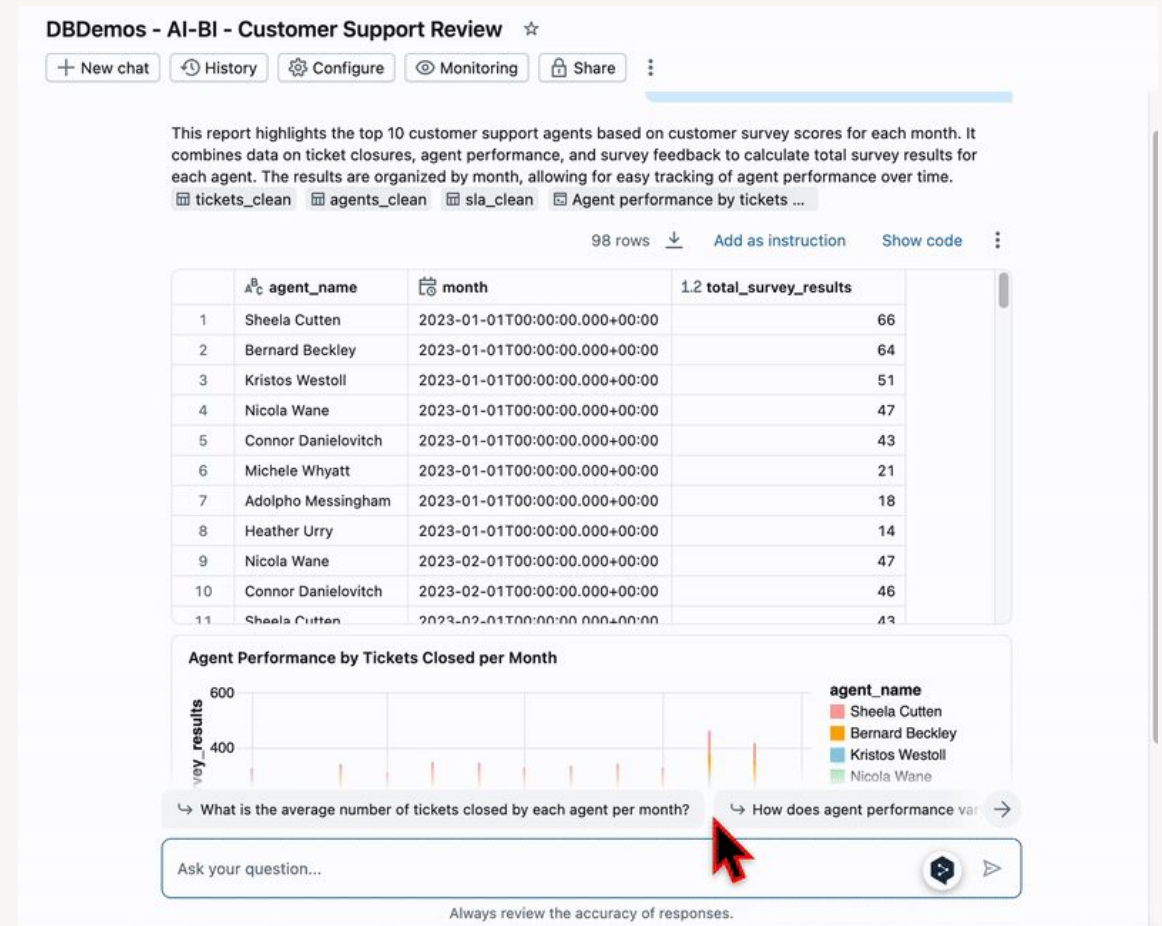
- A built-in feedback system where users can give explicit signals (thumbs up/down, "Request Review," comments) on Genie outputs

Why it matters

- Continuously improves Genie's quality and accuracy by capturing real-world user input and routing it to authors for action

Key Benefits/Use Cases

- Improves model responses with human expertise
- Surfaces gaps in understanding or business logic
- Rapid correction of mistakes and adaptation to business needs



Conversation Context

Session history and state persistence for contextual understanding

What it is

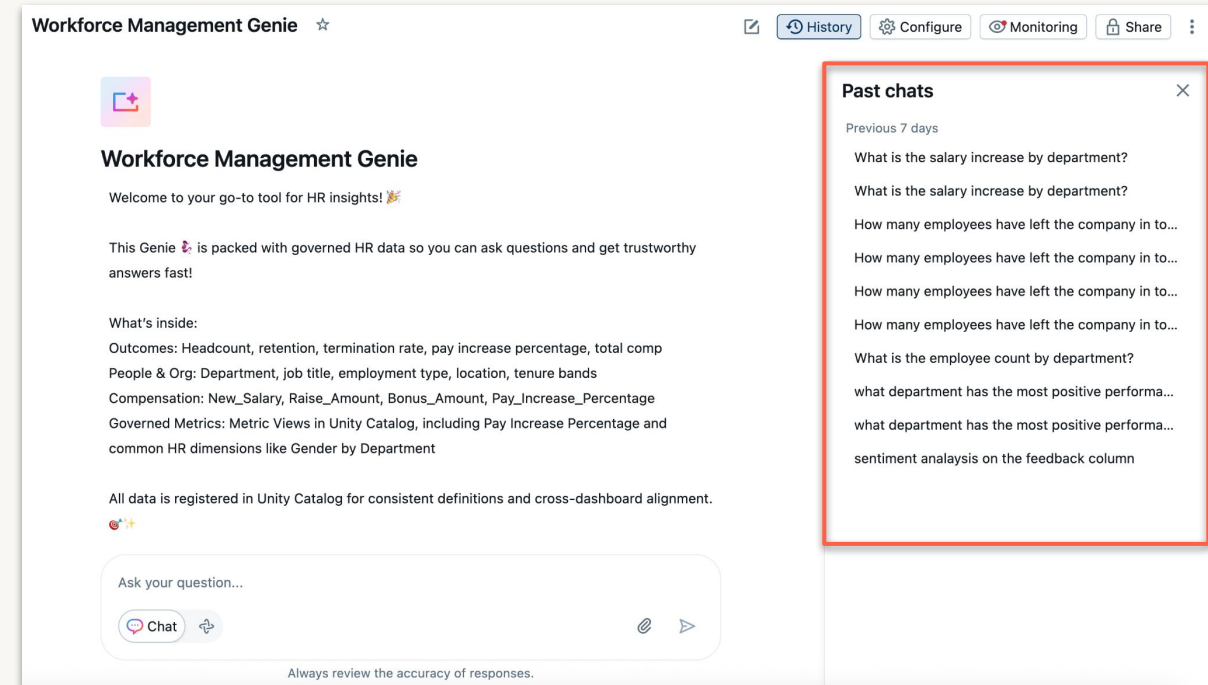
- Genie retains the state and context of ongoing conversations, enabling context-aware Q&A

Why it matters

- Genie can interpret nuanced follow-up questions and adjust responses dynamically based on the user's session history and prior clarifications

Key Benefits/Use Cases

- Session context is continually updated, so Genie's answers reflect everything shared or clarified



Proactive Clarification

Agentic Reasoning in Genie

What it is

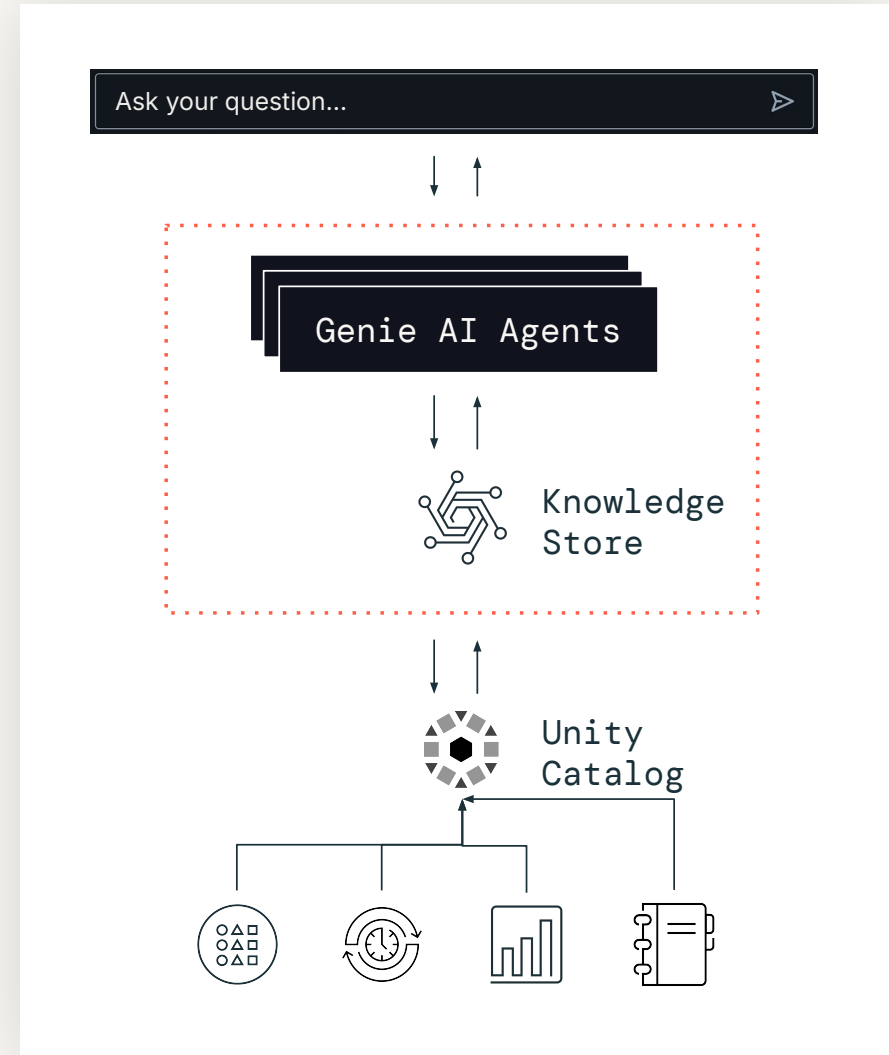
- Genie detects ambiguity or uncertainty in user requests and asks follow-up questions, rather than making low-confidence predictions

Why it matters

- Reduces the risk of incorrect answers and increases user trust by ensuring Genie only responds when intent is clear

Key Benefits/Use Cases

- Clarifications are remembered and shape Genie's future replies within the analytic session



Benchmarking

Measuring the quality and reliability of Genie answers

What it is

- A set of user defined, space-specific questions with known expected results

Why it matters

- Ensure accurate, reliable outputs
- Verify alignment with business logic
- Help build trust in AI-generated insights

Key Benefits/Use Cases

- Detects drift or inconsistencies over time
- Surfaces regressions or bugs early and before org-wide adoption
- Maintains enterprise-grade reliability in natural language analytics

The screenshot displays the 'DBDemos - AI-BI - Customer Support Review' interface with a 'Benchmarks' tab selected. A modal window titled 'Add benchmark' is open, allowing users to define a new benchmark. The modal includes a 'Question' field with the text 'Top agents by number of tickets closed per month' and a small icon. Below this is an 'SQL Answer (optional)' section with a 'Run' button and a text area containing instructions: 'Enter a ground truth SQL statement that correctly answers your benchmark question. This statement will be executed, and the query results will be compared to Genie's results to determine accuracy. Questions without a ground truth SQL statement will be marked for manual review.' At the bottom of the modal are 'Cancel' and 'Add benchmark' buttons. The background interface shows navigation links like '+ New chat', 'History', 'Configure', 'Monitoring', and 'Share', along with '+ Add benchmark' and 'Run benchmarks' buttons.



AI/BI Genie – Governance and Security



End-to-End Security Model

Genie enforces full access controls

What it is

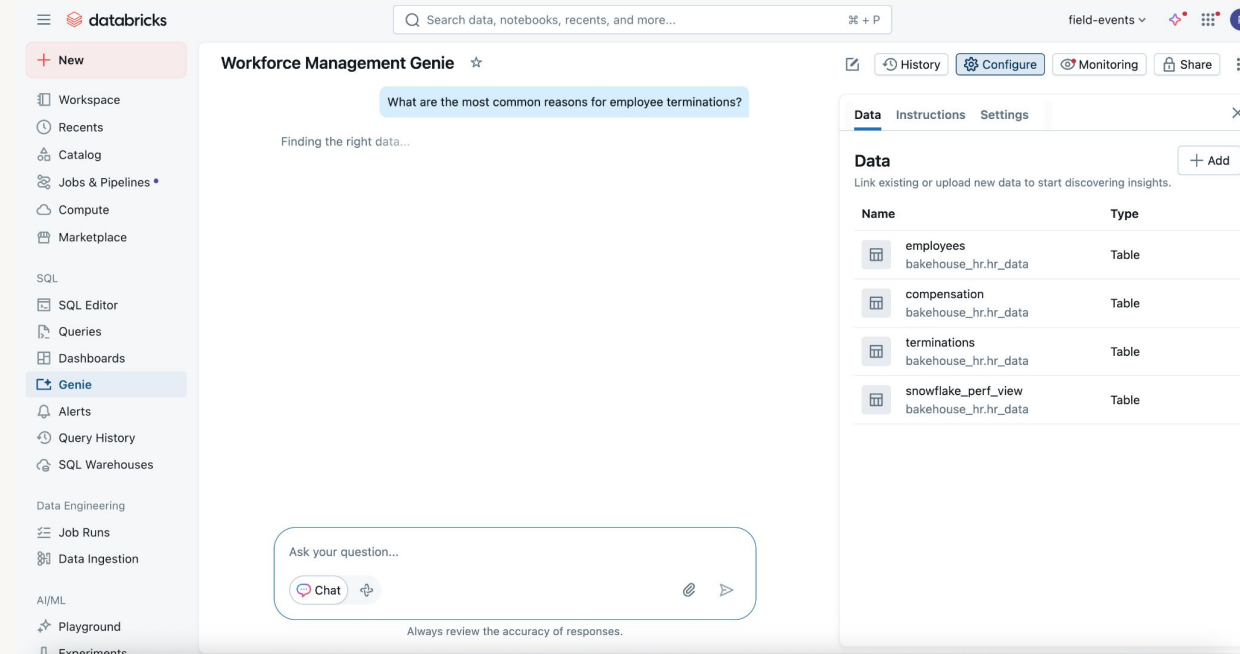
- Every Genie query runs in the exact permission and security context as if the user manually ran SQL in a Databricks notebook, dashboard, etc.

Why it matters

- Guarantees that all data access, filtering, and masking are strictly enforced for each individual

Key Benefits/Use Cases

- Users only see and query data they are entitled to so no risk of accidental overexposure or leaks
- Uniform security compliance across Genie, dashboards, notebooks, and all user interfaces



Audit-Logging

Every question, clarification, and answer

What it is

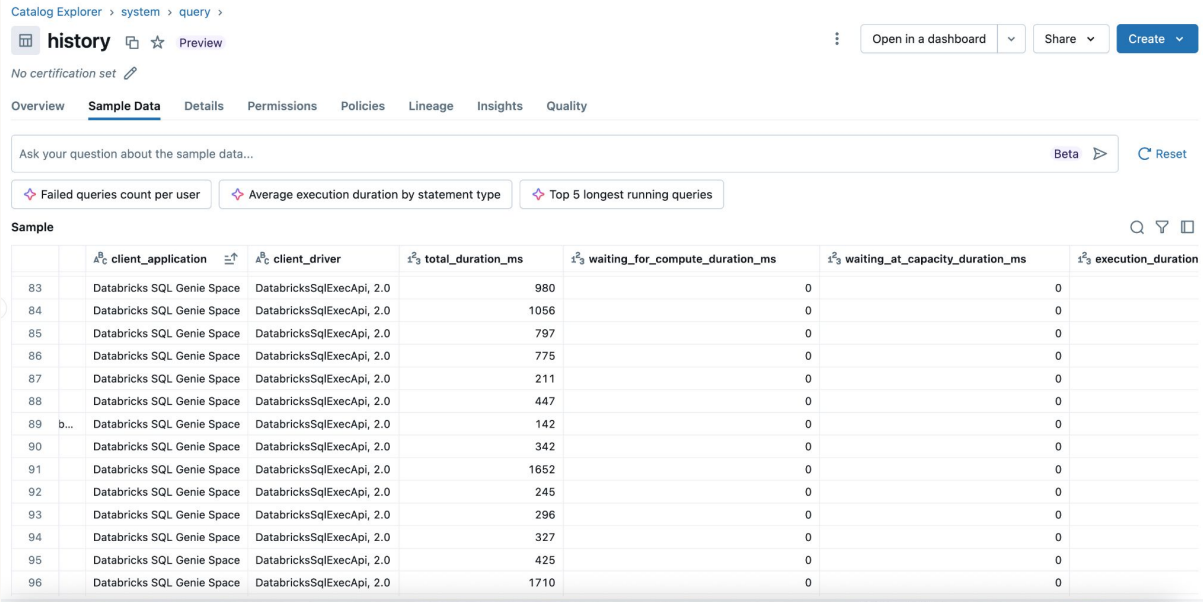
- Genie logs each event with details such as user identity, timestamp, workspace is done through System Tables

Why it matters

- Full transparency, compliance, and traceability for all Genie activities across users and workspaces

Key Benefits/Use Cases

- Supports compliance and governance
- Enables post-hoc review of user queries
- Audits AI behavior for transparency and trust



The screenshot shows the 'history' page in the Databricks Catalog Explorer. The page has a navigation bar with 'Overview', 'Sample Data', 'Details', 'Permissions', 'Policies', 'Lineage', 'Insights', and 'Quality'. The 'Sample Data' tab is active. Below the navigation bar is a search bar with the text 'Ask your question about the sample data...'. To the right of the search bar are buttons for 'Open in a dashboard', 'Share', and 'Create'. Below the search bar are three tabs: 'Failed queries count per user', 'Average execution duration by statement type', and 'Top 5 longest running queries'. The 'Sample Data' tab is selected, and it displays a table with the following columns: 'id', 'client_application', 'client_driver', 'total_duration_ms', 'waiting_for_compute_duration_ms', 'waiting_at_capacity_duration_ms', and 'execution_duration'. The table contains 17 rows of data, with the first row having an 'id' of 83 and a 'total_duration_ms' of 980. The table is titled 'Sample' and has a 'Beta' label and a 'Reset' button.

	client_application	client_driver	total_duration_ms	waiting_for_compute_duration_ms	waiting_at_capacity_duration_ms	execution_duration
83	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	980	0	0	
84	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1056	0	0	
85	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	797	0	0	
86	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	775	0	0	
87	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	211	0	0	
88	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	447	0	0	
89	b...	DatabricksSqlExecApi, 2.0	142	0	0	
90	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	342	0	0	
91	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1652	0	0	
92	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	245	0	0	
93	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	296	0	0	
94	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	327	0	0	
95	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	425	0	0	
96	Databricks SQL Genie Space	DatabricksSqlExecApi, 2.0	1710	0	0	



AI/BI Genie – Usage Patterns & API



Key API Endpoints

Genie Conversation Management

What it is

- Core REST API endpoints to launch, manage, and organize conversations in Genie spaces

Why it matters

- Enables applications and users to programmatically control Genie chat sessions, manage their life cycles, and list accessible Genie workspaces

Key Benefits/Use Cases

- Automate creation and interaction with natural language analytics conversations from any application
- Enables multi-turn conversations with follow-up questions

Functionality	Description	Type	Endpoint
Start conversation	Starts a new chat thread with an initial question	POST	/api/2.0/genie/spaces/{space_id}/start-conversation
Create new conversation message	Ask a follow up question to an existing chat thread	POST	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages
Get conversation message	Track the status of your question + retrieve the generated SQL & description	GET	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}
Get SQL query results	Retrieve the query results from the executed SQL	GET	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}/query-result
Re-execute SQL query	Re-execute SQL that may have expired query results	POST	/api/2.0/genie/spaces/{space_id}/conversations/{conversation_id}/messages/{message_id}/execute-query



AI/BI Genie – Tips, Tricks and Best Practices



Genie: Conceptually

Best practices for Genie space setup

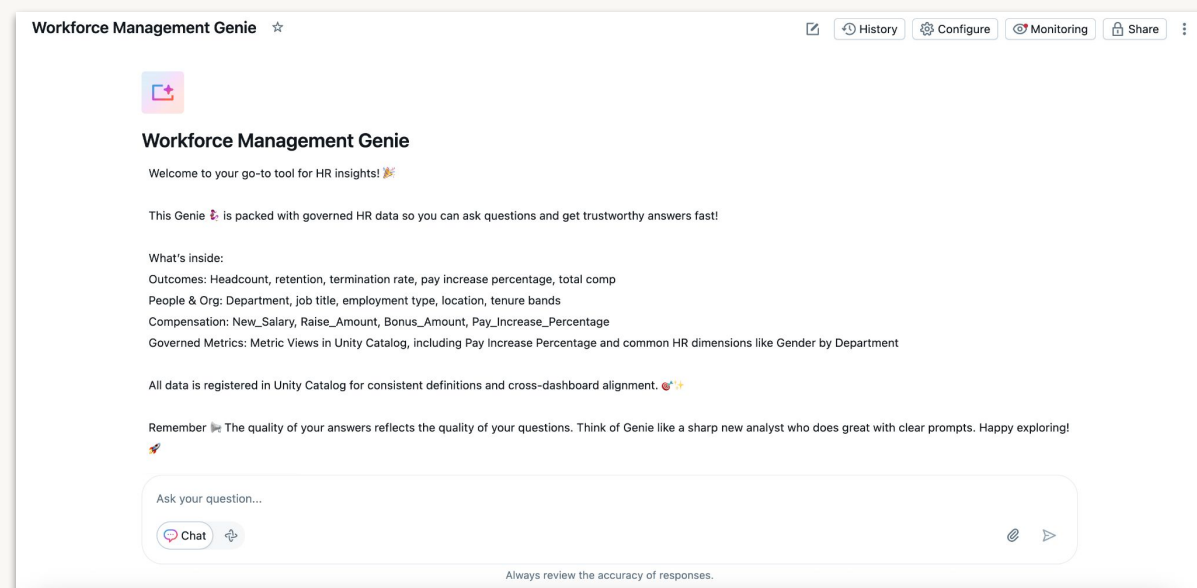
- Think of Genie as a **brand new analyst** joining your organization as their first job post university
- This analyst is going to be **pretty good at writing SQL but will have no pre-existing knowledge about your business** – not high level concepts, not low level jargon
- The only knowledge this analyst will have about your business is the context provided in that particular space
 - **If context is vague or contains conflicting guidance, expect Genie to get confused**, just like a human analyst would be



Genie: Topic selection

Best practices for Genie space setup

- Create a Genie space that maps to a single, focused topic
 - Just like there's no single dashboard for the organization, there's no single Genie space
- A topic should be broad enough to encapsulate a meaningful dataset but narrow enough to provide focused insights
 - Examples include product line performance, marketing campaign analysis, in stock inventory, regional supply chain
- Gather a representative set of 10–15 questions end users typically ask of this data



Genie: Benchmarks

Best practices for Genie space setup

- Create a set of questions that you expect users to ask as benchmarks
- Create “correct SQL” for each benchmark question; it may take some iteration to make sure the question fits the answer
- Run benchmark as you iterate on your instructions

The screenshot displays the Databricks Genie Benchmarks interface. On the left, a list of benchmark questions is shown, with the first one selected: "which mid market accounts have...". The main panel shows the details for this benchmark. The question is "which mid market accounts have generated the most sales globally?". The assessment is "Good". The response section shows the "Model output SQL" and the "Ground truth SQL", both of which are identical. Below the SQL, a table shows the results of the query.

Sales Opportunities > **Benchmarks**

which mid market accounts have generated the most sales globally?

Assessment: Good

Question

which mid market accounts have generated the most sales globally?

Response

Model output SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21



Genie: Data preparation

Best practices for Genie space setup

- **UC metadata is the first and best option** for providing Genie with the context it needs to answer end user questions
- Focused set of UC tables with **defined PK/FK relationships and populated table/column descriptions**
 - Flatten into views if needed, dropping extraneous columns
- Include **value dictionaries** on key string columns

The screenshot displays the Databricks Genie interface within the 'Sales Opportunities' workspace. The 'Benchmarks' tab is active, showing a list of questions on the left. The selected question is 'which mid market accounts have generated the most sales globally?'. The interface shows an assessment of 'Good' and the question text in a search bar. Below, the 'Response' section contains two panels: 'Model output SQL' and 'Ground truth SQL', both displaying identical SQL queries. The queries select account names and total sales, rounded to 2 decimal places, from the 'acme_co_sales.big_rock_schema.accounts' table, joined with 'acme_co_sales.big_rock_schema.big_rock_opportunities' on account ID. The results are filtered by 'o.stagename = '5. Closed Won'' and 'a.company_size_segment__c = 'MM'', grouped by account name, ordered by total sales descending, and limited to 5 results. Below the SQL panels, two tables show the results. Both tables have columns 'account_name' and 'total_sales'. The first table shows 'Pixope' with '189,887.21' in sales. The second table also shows 'Pixope' with '189,887.21' in sales.

Model output SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rock_schema.accounts a
  JOIN acme_co_sales.big_rock_schema.big_rock_opportunities o
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rock_schema.accounts a
  JOIN acme_co_sales.big_rock_schema.big_rock_opportunities o
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment__c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

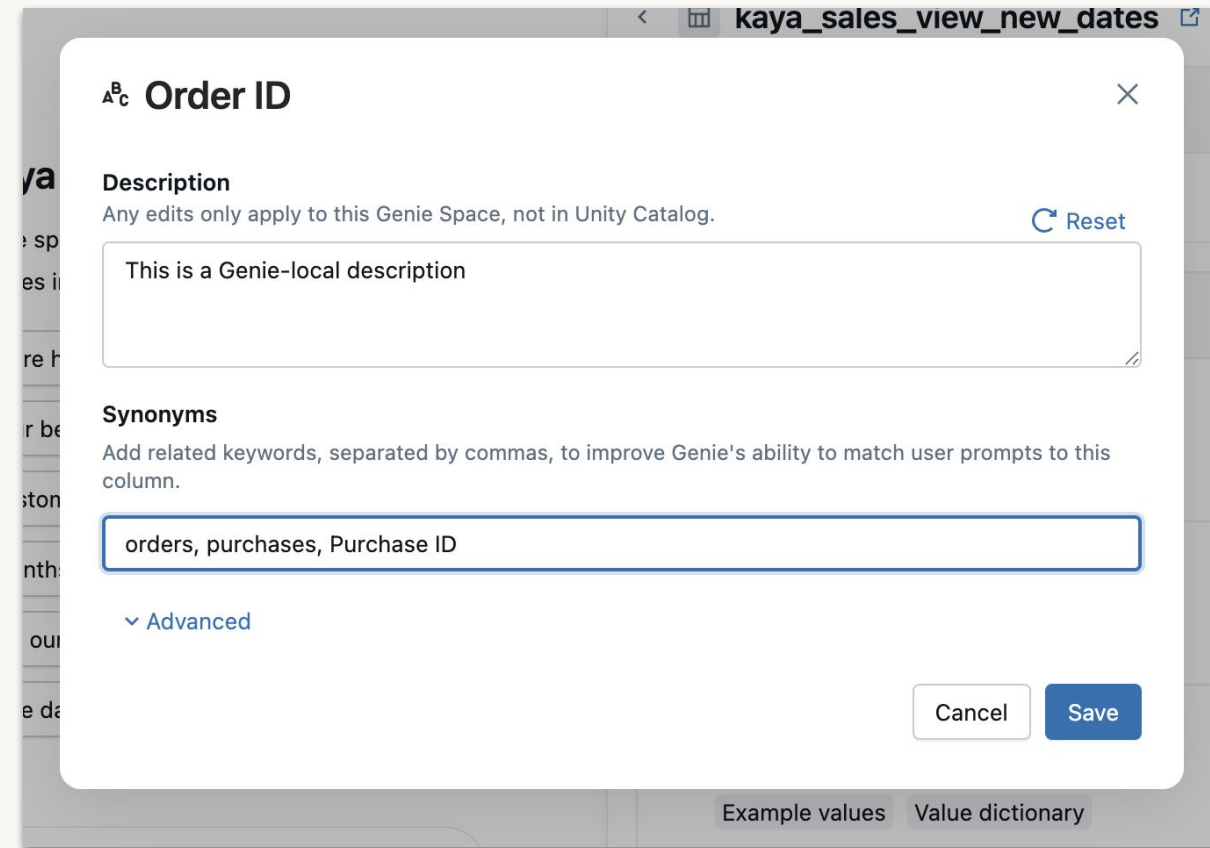
	account_name	total_sales
1	Pixope	189,887.21



Genie: Data preparation

Best practices for Genie space setup

- **Knowledge Store** provides Genie local semantics for when you can't/don't want to update UC metadata
 - Space-level table/column descriptions
 - Column synonyms
 - Show/hide columns
 - Joins
- Include **value dictionaries** on key string columns



The screenshot shows a configuration window for a column named "Order ID". The window has a title bar with a close button (X) and a "Reset" button. The "Description" section contains a text area with the placeholder text "This is a Genie-local description". The "Synonyms" section contains a text area with the text "orders, purchases, Purchase ID". At the bottom right are "Cancel" and "Save" buttons. The background shows a table with columns like "Example values" and "Value dictionary".

Order ID

Description
Any edits only apply to this Genie Space, not in Unity Catalog. [Reset](#)

This is a Genie-local description

Synonyms
Add related keywords, separated by commas, to improve Genie's ability to match user prompts to this column.

orders, purchases, Purchase ID

Advanced

Cancel Save

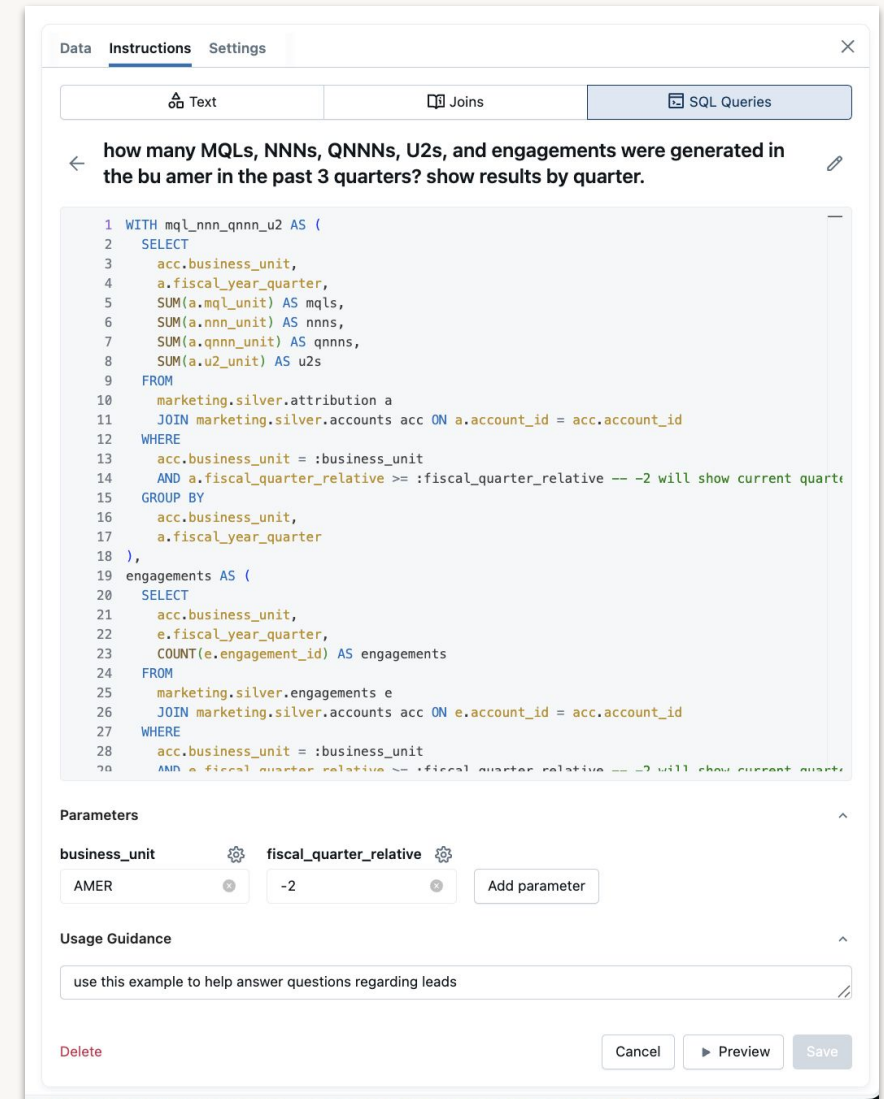
Example values Value dictionary



Genie: Example SQL

Best practices for Genie space setup

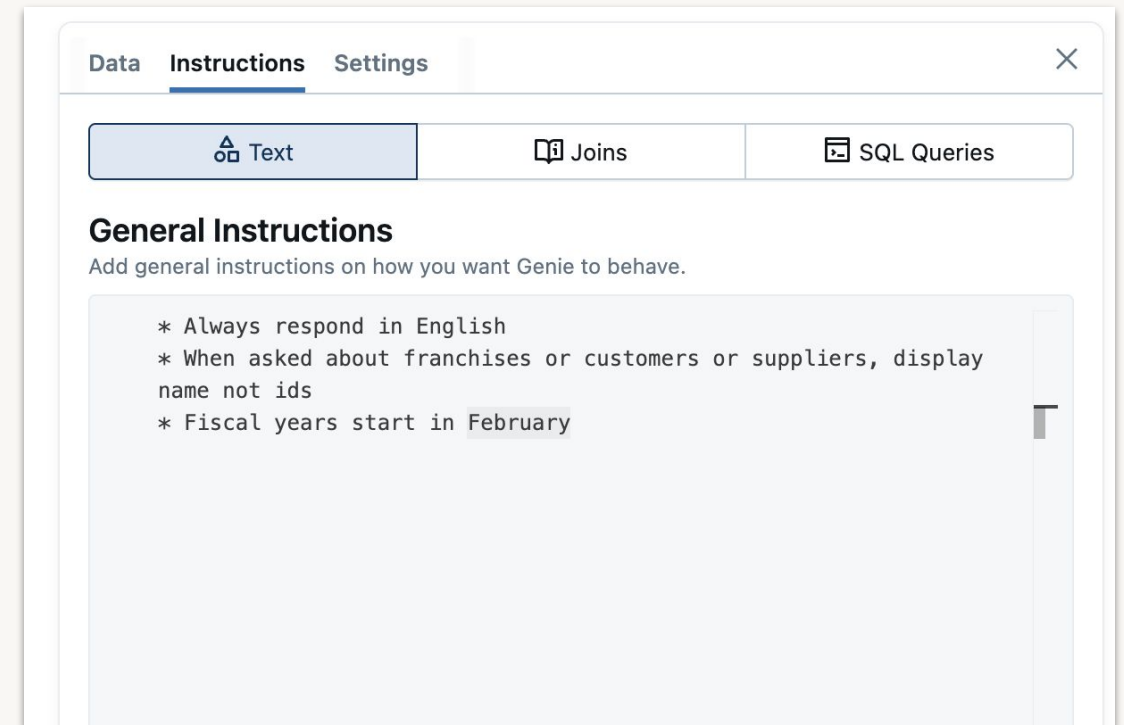
- After UC metadata and knowledge store, this is your **second best option** providing Genie with context
- Leverage **parameterized example SQL** wherever possible to illustrate the representative questions you gathered
- **Leverage Usage Guidance to include additional guidance or alternate phrasing** that should map to that same SQL
 - The question title will have more impact on Genie's behavior than Usage Guidance so include the most common phrasing as the title
- Bias towards Example SQL over UC Functions when possible



Genie: General text instructions

Best practices for Genie space setup

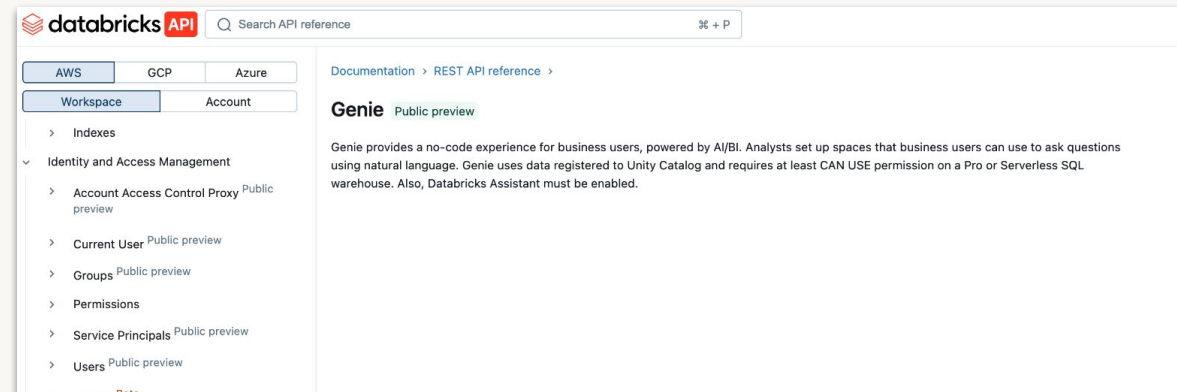
- **General text instructions are the last option** – if something could be expressed as table metadata or Example SQL, it should. E.g.
 - Use example SQL to illustrate how to calculate margin
 - Illustrate join relationships via Knowledge Store Joins
- Best for context that should apply to every single prompt & response, e.g.
 - Format all numeric results to two decimal places
 - All questions related to sales must include a time range that has a specific year; if that is omitted refuse to answer and ask the user to specify
 - Fiscal years start in February
- There is not any filtering on General text instructions so occupies a lot of context



Genie: Application best practices

Best practices for Genie space setup

- REST API documentation
 - Genie conversation API is stateful – conversation history is stored
 - To use in stateless way, start a new conversation for each question
- Implement retry or queuing logic in your application to smooth the user experience during peak concurrency
- Cache answers to common questions to skip Genie API. Doing so will increase determinism and decrease throughput required



Genie: Rollout

Best practices for Genie space setup

- Once you feel your space is ready, **use it yourself for a few weeks to answer new questions** that come in
 - Adjust instructions as needed
- Identify a group of 5–10 initial test users.
Ahead of rollout to that group:
 - Update description & starter questions
 - Establish **benchmarks**
 - Configure **alerts** on user feedback
- Expand to a broader audience after success with the initial group

The screenshot displays the Databricks Genie interface within a workspace named 'Sales Opportunities'. The 'Benchmarks' tab is active, showing a list of questions on the left and a detailed view of a selected benchmark on the right. The selected question is 'which mid market accounts have generated the most sales globally?'. The assessment is 'Good'. The question text is 'which mid market accounts have generated the most sales globally?'. The response section shows the 'Model output SQL' and the 'Ground truth SQL', both of which are identical. Below the SQL, there are two tables showing the results of the query. The first table, 'Model output', has columns 'account_name' and 'total_sales', with one row showing 'Pixope' and '189,887.21'. The second table, 'Ground truth', has the same columns and one row showing 'Pixope' and '189,887.21'.

Sales Opportunities ★ > Benchmarks

+ New chat History Configure Monitoring Share

which mid market accounts have...

top 10 customers this year?

top 10 customers by sales in 2024?

Give me top 10 customers by revenue ...

which mid market accounts have...

Which accounts have we generated th...

Who is my largest customer in europe?

who are my most active sales reps by...

Who is my best rep in asia by sales?

Who are my best 5 salespeople by...

which mid market accounts have generated the most sales globally?

Assessment: Good

Question

which mid market accounts have generated the most sales globally?

Response

Model output SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment_c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21

Ground truth SQL

```
SELECT
  a.name AS account_name,
  ROUND(SUM(o.amount), 2) AS total_sales
FROM
  acme_co_sales.big_rocks_schema.accounts a
  JOIN acme_co_sales.big_rocks_schema.big_rock_opportuni
    ON a.id = o.accountid
WHERE
  o.stagename = '5. Closed Won'
  AND a.company_size_segment_c = 'MM'
GROUP BY
  a.name
ORDER BY
  total_sales DESC
LIMIT 5
```

	account_name	total_sales
1	Pixope	189,887.21



Alert on user feedback

Best practices for Genie space setup

- The audit system table will typically reflect all events, including Genie user feedback, within ~15 minutes
- You can query that system table and then user that query in an alert (documentation)

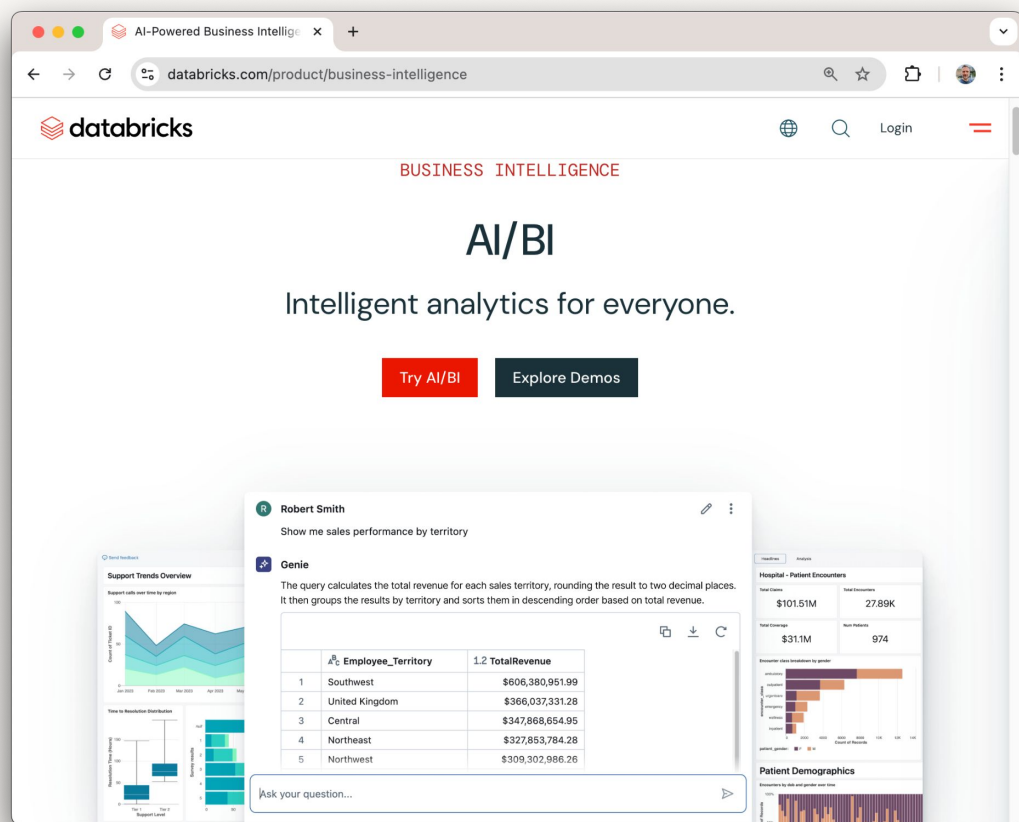
Feedback via API

```
SELECT
  user_identity.email as user_email,
  action_name,
  request_params.space_id,
  request_params.feedback_rating,
  *
FROM
  system.access.audit
WHERE
  service_name = 'aibiGenie'
  AND action_name =
    'genieSendMessageFeedback'
  AND event_date >= current_date() -
    interval 30 days
```



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